Mitigating Risks in Marketplace Semantic Search: A Dataset for Harmful and Sensitive Query Alignment

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Abstract

Semantic search engines have transformed user interaction with online marketplaces, creating a need for effective methods to moderate harmful and sensitive content. Existing approaches often struggle with ambiguous query intent, content classification challenges, and noisy data, making it difficult to ensure user safety while maintaining relevance in search results. To address these challenges, we introduce SHIELD, a synthetic dataset designed for classifying user queries into harmful, sensitive, and normal categories. SHIELD is generated using a large language model with a structured taxonomy, followed by automated filtering using a reward model to ensure data quality and relevance. To demonstrate SHIELD's utility, we evaluate three classification approaches: (1) BM25, a computationally efficient retrieval-based method; (2) a sentence transformer with FAISS, which improves classification by leveraging semantic embeddings; and (3) MoralBERT, a fine-tuned transformer model trained on SHIELD for direct query classification. We discuss the trade-offs among these methods in terms of accuracy, resource requirements, and explainability, highlighting their applicability in real-world semantic search systems. This work provides a foundation for developing AI-driven content moderation systems in semantic search, offering insights into the trade-offs between efficiency, accuracy, and explainability. The SHIELD dataset, pre-trained model, and generation details are publicly available to support future research and real-world deployment: https://github.com/flpspacek/SHIELD. Warning: This paper contains examples of language and content that some readers may find offensive. These examples are included solely to illustrate challenges in content moderation and to highlight the importance of ethical considerations in semantic search systems.

CCS Concepts

• **Information systems → Query intent**; Search interfaces; *Information extraction.*

Keywords

Semantic Search, Recommender Systems, Ethical AI, Dataset Generation, Harmful Content, Sensitive Content

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1 Introduction

Semantic search engines have become essential in online marketplaces, enabling users to find products and services with greater ease and accuracy. These systems rely on advanced natural language understanding to bridge the gap between user intent and available inventory. However, this flexibility introduces significant ethical and operational challenges, particularly in detecting and mitigating harmful or sensitive content. For instance, while a user searching for "something to cut onions with" might appropriately be shown knives, a query seeking "something to stab someone with" should be blocked to protect society and prevent potential public relations disasters. Similarly, sensitive items such as adult toys should not be displayed when a user is searching for children's toys. This dual challenge of preventing inappropriate content exposure while respecting user intent highlights a critical gap in current semantic search systems. Existing solutions often lack the contextual understanding or policy alignment needed to make such fine distinctions, underscoring the need for robust, ethically-aware content moderation mechanisms tailored to semantic search in modern e-commerce systems.

Existing approaches to undesired content detection are not fully suited to the unique challenges posed by semantic search systems. Unlike traditional keyword-based systems, semantic search interprets user intent in a contextual and natural language-driven manner. This complexity introduces challenges in classifying ambiguous or sensitive queries, where the context significantly alters the meaning. For example, a query like "show me something sharp" could suggest kitchen knives in one context but imply harmful intent in another. Current datasets are often not designed to address such context-dependent distinctions or the real-time adaptability required in semantic search environments. As a result, ensuring effective content moderation in semantic search demands datasets and models explicitly tailored for this use case.

To address these challenges, we propose SHIELD (Semantic Harmful-content Identification and Ethical Labeling Dataset), a novel synthetic dataset specifically designed for undesired content detection in semantic search systems. SHIELD combines broad coverage of harmful, sensitive, and safe content with fine-grained taxonomies, enabling robust classification of both user queries and product descriptions. Using this dataset, we trained MoralBERT, a classifier capable of categorizing queries and product descriptions into three categories: harmful, sensitive, and safe. The dataset can

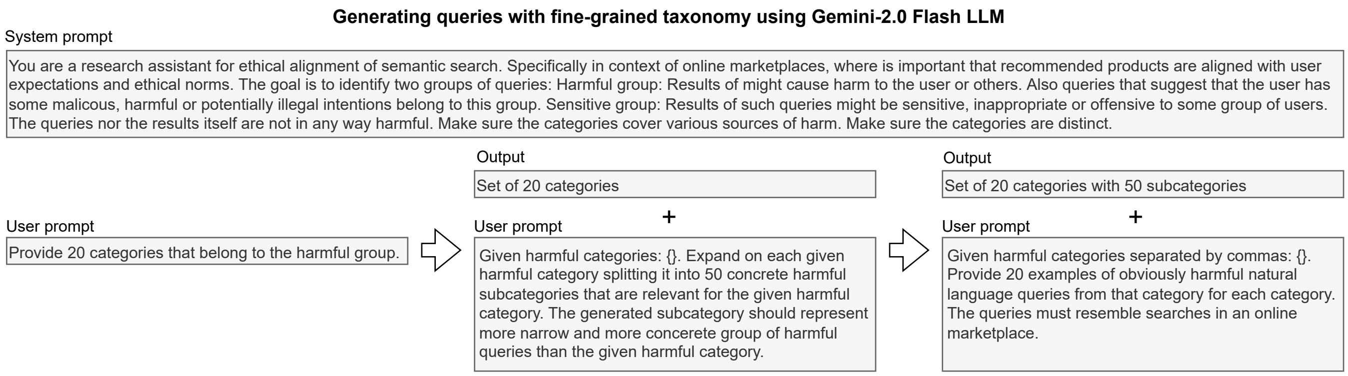

Figure 1: Structured query generation workflow using Gemini-2.0 Flash LLM. The process begins with defining broad categories for harmful and sensitive queries, which are recursively expanded into finer-grained subcategories. Finally, natural language queries are generated for each subcategory, ensuring comprehensive coverage of potential search intents in an online marketplace.

be used directly for similarity-based query filtering or as training data for models like MoralBERT, providing flexibility for integration into various semantic search systems.

Our contributions are as follows:

- We introduce SHIELD, a novel synthetic dataset that provides broad coverage of harmful, sensitive, and safe content with fine-grained taxonomies, addressing critical gaps in existing datasets.
- We train MoralBERT, a high-accuracy model for classifying both content and user queries, ensuring ethical alignment and preventing exposure to inappropriate recommendations in semantic search systems.
- We share SHIELD, the pre-trained MoralBERT model, and detailed dataset generation workflows to foster transparency and enable further advancements in ethical AI.

The rest of the paper is structured as follows: In Section 2, we review related work on undesired content detection and ethical AI. Section 3 describes the generation of SHIELD and the design of its taxonomy. Section 4 outlines the training and evaluation of MoralBERT, including experimental results. Finally, in Section 5, we discuss the broader implications of our work and propose directions for future research.

2 Related Work

The widespread adoption of internet browsing has heightened ethical concerns regarding potential harm to users. These include privacy violations, such as the exposure of queries and personal information, and harmful search results, such as malware or offensive content, which can negatively impact users or compromise their devices. To mitigate these risks, search engines have increasingly adopted machine learning techniques to detect and filter malicious content, thereby enhancing user safety and satisfaction [11, 15]. While modern web browsers incorporate protective mechanisms [1], the lack of a clear consensus on what content should be moderated often leads to conflicts with user expectations [27].

With the rise of social networking platforms, the need for effective content moderation has intensified, as these platforms must balance user safety with free expression. Machine learning systems have been developed to detect and remove hateful or prohibited content [21], with recent work highlighting the multidimensional nature of online harm and the importance of fine-grained, multi-label datasets [4]. Despite these advances, ethical concerns regarding content moderation remain an ongoing area of debate, particularly in terms of transparency, bias, and fairness [26].

Several machine learning competitions in toxic comment classification [6, 7, 14] were held recently resulting in models like ToxicBert [10]. However, ToxicBert fails in detecting harmful queries specific to e-commerce: for example, a query "flasks for cooking crystal meth" gets a score of 0.0003 (resulting in classification as not toxic). However, selling products based on this query is not in the online marketplace provider's best interests, potentially resulting in PR disaster or legal consequences.

The proliferation of generative AI tools, such as ChatGPT, Gemini, Llama, and Stable Diffusion, has introduced new challenges in ethical AI governance. These models, while capable of providing rapid access to aggregated information, raise significant concerns about misinformation, content bias, and harmful outputs [25]. For instance, unaligned models can generate harmful prompts or outputs, as demonstrated by jailbreak attacks in [12]. Mitigation frameworks, such as those proposed in [13], seek to reduce these risks without sacrificing utility. Furthermore, recent work suggests that large language models (LLMs) can themselves be leveraged for content moderation tasks, as shown in [2] or [16].

In parallel, the integration of generative AI with image generation models, such as diffusion models, has underscored the need for ethical safeguards. Without proper alignment, these models risk generating inappropriate or harmful content, including non-consensual explicit imagery, depictions of violence, or politically sensitive material. Such outputs carry significant societal risks, prompting the development of frameworks like Safe Latent Diffusion [24], which restricts the generation of harmful content.

As ethical and responsible AI systems gain momentum across domains such as education, journalism, and e-commerce, ensuring user protection remains a critical priority [9]. Efforts to safeguard users—particularly minors—against objectionable content have led to the development of specialized datasets for conversational AI systems. For example, datasets such as [3] and [17] address toxic prompts and unsafe behaviors, while [19] provides tools for identifying harmful content. However, these datasets are primarily designed for conversational AI and do not align with the unique needs of semantic search systems.

Semantic search, which leverages the capabilities of modern LLMs, remains an underexplored yet crucial area for research. Unlike conversational agents, semantic search engines must address distinct ethical challenges, such as identifying harmful queries and preventing the recommendation of inappropriate or dangerous content. Despite the growing use of AI in search technologies, current research and datasets fail to adequately address the alignment of semantic search systems with ethical guidelines. Advancing this space is essential for developing safe, responsible, and reliable semantic search applications.

3 Dataset Generation Procedure

The first step in creating SHIELD involved generating a structured taxonomy of *harmful*, *sensitive*, and *safe* queries using a **recursive expansion** process with the Gemini-2.0 Flash LLM from Google[1]. Instead of manually defining categories, the model was prompted to autonomously generate high-level topics, which were then expanded into fine-grained subcategories to ensure comprehensive coverage. Each query type—harmful, sensitive, and safe—was handled separately, reflecting distinct ethical and policy considerations. Harmful queries included content promoting illegal, dangerous, or unethical activities (e.g., drug sales, self-harm, fraudulent schemes). Sensitive queries covered topics that were not inherently harmful but could be considered inappropriate or controversial (e.g., adult products, religious items, political merchandise). Safe queries represented common marketplace searches (e.g., apparel, books, home goods).

This process began with generating 20 broad categories for each query type, ensuring coverage across a diverse range of topics. These categories were then recursively expanded into 50 more specific subcategories, creating a structured and detailed taxonomy[2]. This hierarchical approach allowed for fine-grained classification, ensuring that both general and highly specific search intents were represented. By systematically refining the category structure, the dataset captured a wide spectrum of queries, forming a solid foundation for realistic and contextually accurate query generation

Once the taxonomy was established, queries were generated for each subcategory to reflect real-world user searches in an online marketplace. The model was prompted to produce 20 natural language queries per subcategory, ensuring variation in wording and structure. To maintain control over output quality, safety settings were dynamically adjusted—"BLOCK_NONE" was used to allow the generation of harmful queries, while "BLOCK_LOW_AND_ABOVE" filtered sensitive content. This approach ensured that SHIELD captured a diverse range of real-world queries, balancing scalability, adaptability, and ethical oversight in semantic search moderation.

While the initial query generation step provided a broad and diverse dataset, not all generated queries were suitable for training a robust moderation system. Some queries lacked clear alignment with their assigned category, while others exhibited inconsistencies in phrasing or intent. To refine the dataset and ensure high-quality, well-classified queries, a structured filtering process was applied using the Skywork Reward Model [18][3]. This model evaluated each query to determine whether it aligned correctly with its classification—harmful, sensitive, or safe. By leveraging this automated scoring system, we systematically removed ambiguous or incorrectly categorized queries while preserving the diversity necessary for robust content moderation.

After scoring all queries, a thresholding mechanism was applied to systematically filter out misclassified or low-confidence queries. To determine an appropriate cutoff, we analyzed the score distributions of safe queries, using them as a baseline for distinguishing harmful and sensitive queries. The 95th percentile of safe query scores was selected as the filtering threshold, serving as a reference point for identifying well-aligned queries. Harmful and sensitive queries scoring above this threshold were retained in their respective categories, while those scoring below were discarded. This approach ensured that only high-confidence, clearly aligned queries remained in the dataset, while ambiguous or low-quality examples were filtered out. Before filtering, the dataset contained approximately 20,000 queries per category; after applying the threshold, 8,871 sensitive and 17,170 harmful queries remained. This statistical approach ensured removing low-quality and ambiguous queries while preserving the diversity necessary for effective semantic search moderation. For examples of generated queries, see Table 1.

4 Dataset Use

The SHIELD dataset is designed for the classification of user queries into three categories: harmful, sensitive, and normal. In this section, we evaluate the effectiveness of three different approaches for utilizing SHIELD in query classification tasks. These approaches include (1) using BM25 [23] for query similarity, (2) leveraging sentence embeddings [22] with FAISS [8] for nearest neighbor search, and (3) fine-tuning a transformer-based model. By comparing these methods, we aim to assess their accuracy, practicality, and applicability in real-world semantic search scenarios.

The SHIELD dataset was randomly split into training, validation, and test sets with proportions of 80%, 10%, and 10%, respectively. All source code for reproducing our experiments is publicly available on our GitHub page[4].

The first classification approach utilizes BM25, a ranking function commonly used in information retrieval to measure the relevance of documents to a query[5]. For each user query in the test set, BM25 is used to compute similarity scores between the query

[1]This particular model's API offers users fine control over safety settings, including the option to disable safety filters regardless of the probability of generating unsafe content. This capability made the Gemini 2.0 Flash model an ideal choice for constructing the dataset, which, by design, incorporates harmful queries and topics.

[2]Some subcategories were removed due to redundancy.

[3]Although larger and more robust models, such as NVIDIA's nemotron-4-340b-reward [20], are available via API, we were unable to gain access to them for this work.

[4]https://github.com/flpspacek/SHIELD

[5]We implement this method using the rank_bm25 [5] Python library

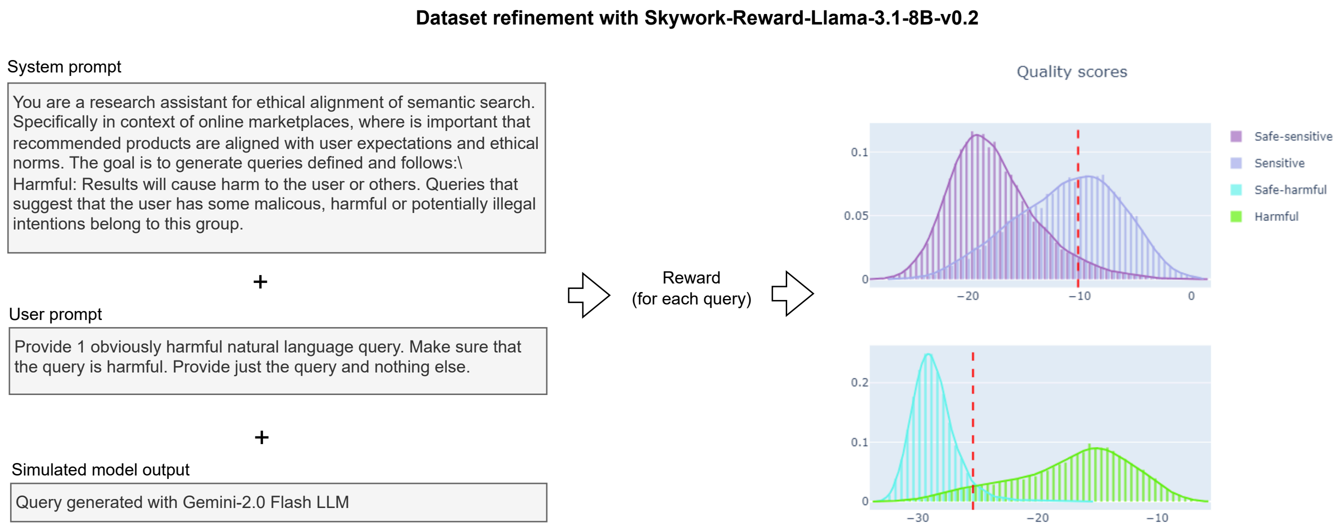

Figure 2: Dataset refinement process begins with Skywork-Reward-Llama-3.1-8B-v0.2 generating a quality score for each query (left), followed by statistical analysis (right). The top graph compares the quality of sensitive queries to safe ones, while the bottom graph shows the quality of harmful queries relative to safe ones. The dashed line indicates the 95th percentile of safe query quality, used as the filtering threshold.

Table 1: Examples of generated queries in the SHIELD dataset

HARMFUL:
glass tubing for meth production
LSD discreet shipping
where can i buy an assault rifle no questions asked
cheap child sexual abuse pictures
Lethal poisonous substances for sale

SENSITIVE:
best vibrator for women
body paint for nude photography
black leather lingerie
online gambling membership offers
Pack of Cuban cigars

SAFE:
hard graphite pencils for technical drawing
kitchen island with stainless steel top
R&B music for date night
professional video editing software
buy kids vr games

and all queries in the training set. The query with the highest similarity score is selected as the closest match, and its label—harmful, sensitive, or normal—is assigned to the test query.

The second classification approach leverages semantic similarity rather than keyword-based matching. We generate dense vector embeddings for each query in both the training and test set with an encoder-based Transformer model[6]. To efficiently search for the most similar query, we use FAISS [8], a scalable nearest neighbor retrieval library. For each test query, FAISS identifies the most similar query from the training set based on cosine similarity in the embedding space. The test query is then assigned the label of its closest match. This approach enhances contextual understanding and generalization over lexical-based methods like BM25.

The third classification approach utilizes MoralBERT, a fine-tuned transformer-based model designed specifically for query classification. We start with ModernBERT [28][7] as the base model and extend it by adding a classification layer. The model is then fine-tuned on the SHIELD training set using supervised learning. MoralBert was trained on 4xV100 GPU server; it takes about 5 minutes to train the model. Unlike BM25 and the sentence transformer approach, which rely on similarity to existing queries, MoralBERT learns a direct classification function, allowing it to generalize beyond previously seen queries. This approach offers the highest accuracy among the three methods but requires more computational resources.

To establish baselines, we evaluated ToxicBERT and the Gemini-2.0 Flash LLM as standalone classifiers for query categorization.

To assess the effectiveness of each approach, we measured accuracy on the SHIELD test set. ToxicBERT achieved an accuracy of 0.5535, a result that aligns with expectations, as ToxicBERT is trained for toxic comment detection rather than identifying harmful

[6]We use the https://huggingface.co/sentence-transformers/multi-qa-mpnet-base-cos-v1 model.
[7]We share the resulting model with sample use at our hugginface repository https://huggingface.co/spacefi1/moralBERT

Table 2: Results of query classification task on the SHIELD dataset.

	Accuracy	F1-score
ToxicBERT	0.55351	0.07091
Gemini LLM	0.80292	0.82785
SHIELD BM25	0.93209	0.93211
SHIELD semantic similarity	0.96497	0.96506
MoralBERT	0.98380	0.98624

intent in the context of product search. In contrast, direct classification using Gemini-2.0 Flash LLM yielded an accuracy of 0.8029, highlighting the added value of our dataset curated using the Skywork-Reward-Llama-3.1-8B-v0.2 model. However, employing a large LLM for real-time classification is likely to be cost-prohibitive, especially given the high query volumes typical in production search systems.

Among the methods leveraging the SHIELD dataset, the BM25 method achieved an accuracy of 93.21%, while the semantic similarity approach using sentence transformers improved performance to 96.49%. The best results were obtained using MoralBERT, our fine-tuned transformer model, which reached an accuracy of 98.38%. A detailed comparison of all evaluated methods is presented in Table 2.

5 Conclusions and future work

In this work, we introduced SHIELD, a synthetic dataset designed for classifying user queries into three categories: harmful, sensitive, and normal. This classification is tailored specifically for semantic search systems operating within the e-commerce domain, where understanding and moderating user intent is critical to maintaining user safety, legal compliance, and brand integrity. The dataset was constructed using a structured generation process powered by the Gemini-2.0 Flash LLM, followed by automated quality filtering to ensure high relevance and accuracy. To further refine the data, we employed the Skywork-Reward model to score and filter queries based on alignment with their intended classification, removing ambiguous or miscategorized entries.

While SHIELD presents a strong foundation for semantic moderation, it is important to note that no human validation was employed in the dataset creation process. As such, we strongly recommend involving human reviewers before deploying this dataset or derived models in production environments. Additionally, practitioners intending to use our approach should consider two key factors. First, domain specificity: different product catalogs or business contexts may require generating a customized dataset using our procedure, tailored to their own marketplace and user behavior. Second, cultural variation: social norms regarding what constitutes harmful or sensitive content can vary significantly across regions. For example, queries involving LGBT topics that are acceptable in many liberal EU countries may be considered inappropriate or even illegal in parts of the Middle East, while political sensitivities may differ between North America and China. We therefore advise users to carefully examine and validate the dataset's behavior within the context of their specific use case and cultural setting.

To demonstrate a possible usage of SHIELD, we explored three classification approaches, each offering distinct trade-offs. BM25 is a highly efficient method that requires minimal computational resources, making it an attractive choice for lightweight systems. However, its reliance on exact keyword matches makes it less effective for handling semantically diverse queries. The sentence transformer with FAISS improves classification accuracy by leveraging semantic embeddings, offering a balance between performance and interpretability, particularly for organizations already utilizing embedding-based retrieval systems. Finally, MoralBERT, a fine-tuned transformer model, achieves the highest accuracy by learning complex patterns directly from data, but at the cost of increased computational requirements. These findings highlight the trade-offs between efficiency, interpretability, and accuracy, providing insights into selecting an appropriate classification method for ethical query moderation in semantic search systems.

While SHIELD provides a strong foundation for query classification in semantic search, several areas remain for future exploration. Expanding the dataset is a key priority, including the addition of more nuanced categories, multilingual queries, and real-world user-generated data to improve diversity and adaptability across different marketplaces. Additionally, integrating context-aware query classification, where a query's meaning is influenced by its surrounding interactions, could further enhance the dataset's applicability.

Beyond dataset improvements, future research should explore fine-tuning and robustness in MoralBERT to mitigate potential vulnerabilities. Investigating adversarial attacks or biases in model predictions will be crucial for ensuring robustness against query manipulation. Another important direction is integration with production systems, where hybrid approaches combining retrieval-based models (sentence transformer) and deep learning classifiers (MoralBERT) could balance efficiency and accuracy.

To validate SHIELD's real-world effectiveness, conducting a user study in an online environment will be essential. This study would measure the impact of SHIELD-based query moderation on user experience, search relevance, and marketplace compliance with content policies. Finally, ethical considerations should remain central to future work, with a focus on fairness, bias mitigation, and adapting content moderation policies to various cultural and regulatory contexts.

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References

[1] 2020. Information quality & content moderation. https://storage.googleapis.com/gweb-uniblog-publish-prod/documents/Info_quality_content_moderation-whie_paper-UPDATED.pdf

[2] Nouar AlDahoul, Myles Joshua Toledo Tan, Harishwar Reddy Kasireddy, and Yasir Zaki. 2024. Advancing Content Moderation: Evaluating Large Language Models for Detecting Sensitive Content Across Text, Images, and Videos. arXiv:2411.17123 [cs.CV] https://arxiv.org/abs/2411.17123

[3] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. 2022. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. arXiv:2204.05862 [cs.CL] https://arxiv.org/abs/2204.05862

[4] Federico Bianchi, Stefanie Anja Hills, Patricia Rossini, Dirk Hovy, Rebekah Tromble, and Nava Tintarev. 2022. "It's Not Just Hate": A Multi-Dimensional Perspective on Detecting Harmful Speech Online. arXiv:2210.15870 [cs.CL] https://arxiv.org/abs/2210.15870

[5] Dorian Brown. 2020. *Rank-BM25: A Collection of BM25 Algorithms in Python*. https://doi.org/10.5281/zenodo.4520057

[6] cjadams, Daniel Borkan, inversion, Jeffrey Sorensen, Lucas Dixon, Lucy Vasserman, and nithum. 2019. Jigsaw Unintended Bias in Toxicity Classification. https://kaggle.com/competitions/jigsaw-unintended-bias-in-toxicity-classification. Kaggle.

[7] cjadams, Jeffrey Sorensen, Julia Elliott, Lucas Dixon, Mark McDonald, nithum, and Will Cukierski. 2017. Toxic Comment Classification Challenge. https://kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge. Kaggle.

[8] Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2024. The Faiss library. (2024). arXiv:2401.08281 [cs.LG]

[9] Philipp Hacker, Andreas Engel, and Marco Mauer. 2023. Regulating ChatGPT and other Large Generative AI Models. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency* (Chicago, IL, USA) *(FAccT '23)*. Association for Computing Machinery, New York, NY, USA, 1112–1123. https://doi.org/10.1145/3593013.3594067

[10] Laura Hanu and Unitary team. 2020. Detoxify. Github. https://github.com/unitaryai/detoxify.

[11] Weiming Hu, Ou Wu, Zhouyao Chen, Zhouyu Fu, and Steve Maybank. 2007. Recognition of Pornographic Web Pages by Classifying Texts and Images. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 29, 6 (2007), 1019–1034. https://doi.org/10.1109/TPAMI.2007.1133

[12] Yangsibo Huang, Samyak Gupta, Mengzhou Xia, Kai Li, and Danqi Chen. 2023. Catastrophic Jailbreak of Open-source LLMs via Exploiting Generation. arXiv:2310.06987 [cs.CL] https://arxiv.org/abs/2310.06987

[13] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabsa. 2023. Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations. arXiv:2312.06674 [cs.CL] https://arxiv.org/abs/2312.06674

[14] Ian Kivlichan, Jeffrey Sorensen, Julia Elliott, Lucy Vasserman, Martin Görner, and Phil Culliton. 2020. Jigsaw Multilingual Toxic Comment Classification. https://kaggle.com/competitions/jigsaw-multilingual-toxic-comment-classification. Kaggle.

[15] P.Y. Lee, S.C. Hui, and A.C.M. Fong. 2002. Neural networks for web content filtering. *IEEE Intelligent Systems* 17, 5 (2002), 48–57. https://doi.org/10.1109/MIS.2002.1039832

[16] Lingyao Li, Lizhou Fan, Shubham Atreja, and Libby Hemphill. 2024. "HOT" ChatGPT: The Promise of ChatGPT in Detecting and Discriminating Hateful, Offensive, and Toxic Comments on Social Media. *ACM Trans. Web* 18, 2, Article 30 (March 2024), 36 pages. https://doi.org/10.1145/3643829

[17] Zi Lin, Zihan Wang, Yongqi Tong, Yangkun Wang, Yuxin Guo, Yujia Wang, and Jingbo Shang. 2023. ToxicChat: Unveiling Hidden Challenges of Toxicity Detection in Real-World User-AI Conversation. arXiv:2310.17389 [cs.CL] https://arxiv.org/abs/2310.17389

[18] Chris Yuhao Liu, Liang Zeng, Jiacai Liu, Rui Yan, Jujie He, Chaojie Wang, Shuicheng Yan, Yang Liu, and Yahui Zhou. 2024. Skywork-Reward: Bag of Tricks for Reward Modeling in LLMs. arXiv:2410.18451 [cs.AI] https://arxiv.org/abs/2410.18451

[19] Todor Markov, Chong Zhang, Sandhini Agarwal, Tyna Eloundou, Teddy Lee, Steven Adler, Angela Jiang, and Lilian Weng. 2022. A Holistic Approach to Undesired Content Detection. *arXiv preprint arXiv:2208.03274* (2022).

[20] Nvidia, :, Bo Adler, Niket Agarwal, Ashwath Aithal, Dong H. Anh, Pallab Bhattacharya, Annika Brundyn, Jared Casper, Bryan Catanzaro, Sharon Clay, Jonathan Cohen, Sirshak Das, Ayush Dattagupta, Olivier Delalleau, Leon Derczynski, Yi Dong, Daniel Egert, Ellie Evans, Aleksander Ficek, Denys Fridman, Shaona Ghosh, Boris Ginsburg, Igor Gitman, Tomasz Grzegorzek, Robert Hero, Jining Huang, Vibhu Jawa, Joseph Jennings, Aastha Jhunjhunwala, John Kamalu, Sadaf Khan, Oleksii Kuchaiev, Patrick LeGresley, Hui Li, Jiwei Liu, Zihan Liu, Eileen Long, Ameya Sunil Mahabaleshwarkar, Somshubra Majumdar, James Maki, Miguel Martinez, Maer Rodrigues de Melo, Ivan Moshkov, Deepak Narayanan, Sean Narenthiran, Jesus Navarro, Phong Nguyen, Osvald Nitski, Vahid Noroozi, Guruprasad Nutheti, Christopher Parisien, Jupinder Parmar, Mostofa Patwary, Krzysztof Pawelec, Wei Ping, Shrimai Prabhumoye, Rajarshi Roy, Trisha Saar, Vasanth Rao Naik Sabavat, Sanjeev Satheesh, Jane Polak Scowcroft, Jason Sewall, Pavel Shamis, Gerald Shen, Mohammad Shoeybi, Dave Sizer, Misha Smelyanskiy, Felipe Soares, Makesh Narsimhan Sreedhar, Dan Su, Sandeep Subramanian, Shengyang Sun, Shubham Toshniwal, Hao Wang, Zhilin Wang, Jiaxuan You, Jiaqi Zeng, Jimmy Zhang, Jing Zhang, Vivienne Zhang, Yian Zhang, and Chen Zhu. 2024. Nemotron-4 340B Technical Report. arXiv:2406.11704 [cs.CL] https://arxiv.org/abs/2406.11704

[21] Amir H. Razavi, Diana Inkpen, Sasha Uritsky, and Stan Matwin. 2010. Offensive Language Detection Using Multi-level Classification. In *Advances in Artificial Intelligence*, Atefeh Farzindar and Vlado Kešelj (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 16–27.

[22] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. https://arxiv.org/abs/1908.10084

[23] Stephen E Robertson, Steve Walker, Susan Jones, Micheline M Hancock-Beaulieu, Mike Gatford, et al. 1995. Okapi at TREC-3. *Nist Special Publication Sp* 109 (1995), 109.

[24] Patrick Schramowski, Manuel Brack, Björn Deiseroth, and Kristian Kersting. 2023. Safe Latent Diffusion: Mitigating Inappropriate Degeneration in Diffusion Models. arXiv:2211.05105 [cs.CV] https://arxiv.org/abs/2211.05105

[25] Bernd Carsten Stahl and Damian Eke. 2024. The ethics of ChatGPT – Exploring the ethical issues of an emerging technology. *International Journal of Information Management* 74 (2024), 102700. https://doi.org/10.1016/j.ijinfomgt.2023.102700

[26] Maronikolakis A. & Wisiorek A. Udupa, S. 2023. Ethical scaling for content moderation: Extreme speech and the (in)significance of artificial intelligence. *Big Data & Society, 10(1)* (2023). https://doi.org/10.1177/20539517231172424

[27] Aleksandra Urman, Aniko Hannak, and Mykola Makhortykh. 2023. User Attitudes to Content Moderation in Web Search. arXiv:2310.03458 [cs.CY] https://arxiv.org/abs/2310.03458

[28] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Nathan Cooper, Griffin Adams, Jeremy Howard, and Iacopo Poli. 2024. Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference. arXiv:2412.13663 [cs.CL] https://arxiv.org/abs/2412.13663